MOS STRUCTURE, CHARGE STORAGE & FLASH MEMORY DEVICES

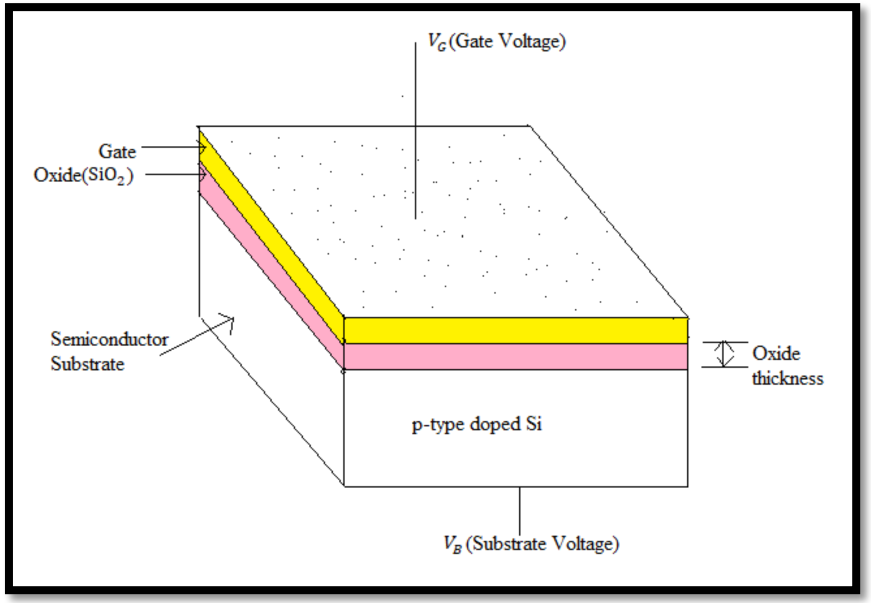
Dr. Archana Sharma

MOS Transistor

- The MOS field effect transistor (MOSFET) is the fundamental building block of MOS and CMOS digital integrated circuits.
- Compared to the bipolar junction transistor (BJT), the MOS transistor occupies a relatively smaller Si area, and its fabrication used to involve fewer processing steps.
- The technical advantages, together with the relative simplicity of MOSFET operation, have helps make the MOS transistor the most widely switching device in LSI and VLSI circuits.

Metal Oxide Semiconductor (MOS) Structure

- The MOS structure consists of three layers.
- The metal gate electrode, the insulating oxide $(\mathrm{Si}O_2)$ layer, and the p-type bulk semiconductor (Si) called the substrate.
- The MOS structure forms a capacitor with the gate and the substrate acting as the two terminals (plates) and the oxide layer as the dielectric.
- The thickness of the $\mathrm{Si}O_2$ layer is usually between 10nm and 50nm.



Working of MOS Structure

When a voltage is applied to the **gate terminal**, it controls the charge distribution in the semiconductor near the oxide interface.

Depending on the applied gate voltage, three operating conditions occur:

(a) Accumulation

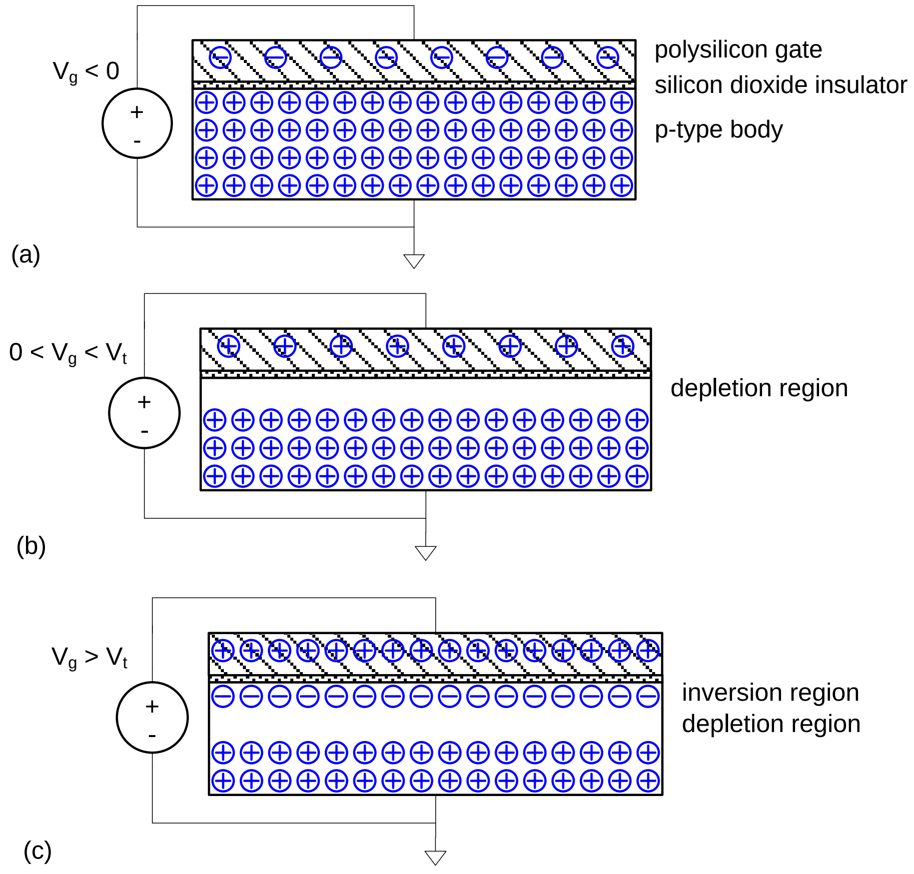
- Occurs when gate voltage attracts **majority carriers**
- For p-type substrate, negative gate voltage attracts holes
- Charge density near the surface increases

(b) Depletion

- Moderate positive gate voltage repels majority carriers
- A depletion region forms near the surface
- Region is depleted of free charge carriers

(c) Inversion

- High positive gate voltage attracts **minority carriers**
- For p-type substrate, electrons form an inversion layer
- This layer forms a **conducting channel**



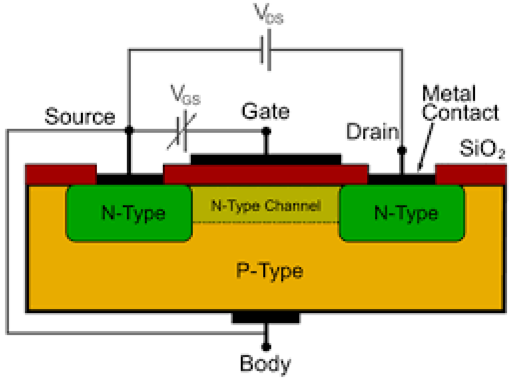
Applications of MOS Transistors

MOS transistors are employed in a vast array of applications across various fields of electronics:

1. **Digital Logic Circuits: MOSFETs are fundamental in digital circuits, particularly in CMOS technology, which is widely used in microprocessors, memory chips, and digital signal processors. Their ability to switch rapidly between on and off states makes them ideal for high-speed logic operations.**
2. **Analog Circuits: In linear applications, MOSFETs serve as amplifiers and switches. They can achieve high gain and low noise, making them suitable for audio and radio-frequency amplification.**
3. **Power Electronics: Power MOSFETs are designed to handle high voltage and current levels used in power supply circuits, motor drivers, and electronic speed controllers.**
4. **RF and Microwave Applications: MOSFETs are utilized in RF amplifiers and mixers due to their high frequency performance and low noise characteristics.**
5. **Sensors and Actuators: In MEMS, MOSFETs are often used as drivers and switches for sensors and actuators, enabling a wide range of applications in automotive and consumer electronics.**

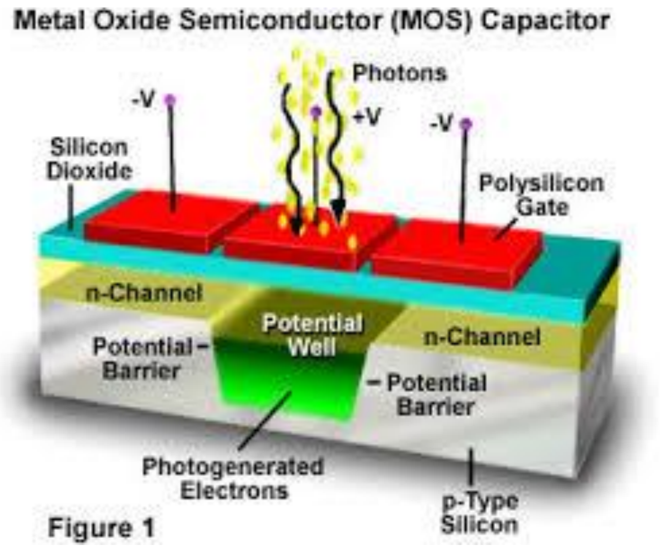
Charge Storage Mechanism

- Charges accumulate at the oxide–semiconductor interface.
- The oxide layer prevents leakage, allowing charge retention
- Stored charge depends on:
  - Applied gate voltage
  - Oxide thickness
  - Semiconductor doping



- The stored charge can consist of mobile carriers in the semiconductor as well as fixed or trapped charges within the oxide or at the oxide–semiconductor interface.
- Charge storage in a MOS structure is purely electrostatic and is governed by the relation Q = CV.

This controlled charge storage capability makes the MOS structure extremely useful in electronic applications, particularly in MOSFETs and memory devices such as DRAM and Flash memory



Importance of Charge Storage

- Used in memory devices
- Enables information storage in digital form
- Basis of non-volatile memory technologies

Due to its low power consumption, high input impedance, scalability, and compatibility with integrated circuit technology, MOS technology forms the backbone of modern microelectronics and VLSI systems

Flash Memory Device

Flash memory is a non-volatile memory that can retain data for extended periods without current supply, with typical data retention spanning 10 years or more. Its storage characteristics are equivalent to solid-state drives (SSDs), making it the foundation for flash memory to become the primary storage medium for various portable digital devices.

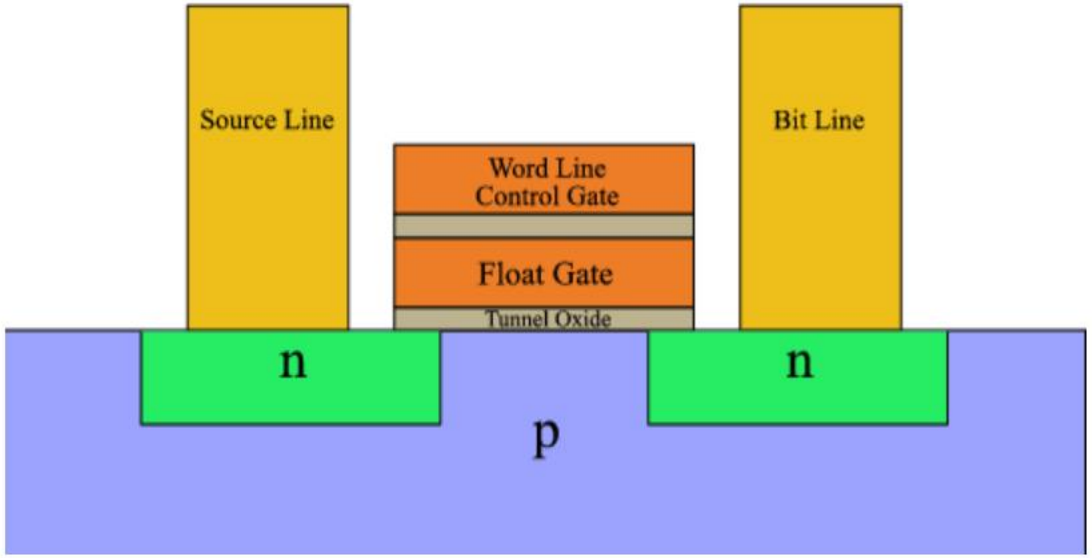
Flash memory offers significant advantages including small size, low power consumption, shock resistance, and immunity to physical damage.

It is an ideal storage medium for mobile digital products. The core memory of all USB flash drives, SSDs, and memory cards in various digital devices is flash memory

Structure of Flash Memory Cell

A flash memory cell is based on a Floating Gate MOSFET, which consists of:

- Control Gate
- Floating Gate (completely surrounded by oxide)
- Source and Drain
- Semiconductor Substrate



Advantages of Flash Memory

- Non-volatile storage
- High speed operation
- Low power consumption
- High storage density
- Long data retention

Applications

- USB flash drives
- Solid State Drives (SSD)
- Memory cards
- Embedded systems
- Smartphones and tablets

Relation Between MOS Structure and Flash Memory

- MOS structure provides **charge control**
- Charge storage enables **data retention**
- Floating gate MOSFET uses MOS principles
- Flash memory is an advanced application of MOS technology

Conclusion

- **The MOS structure is the foundation of modern electronics.**
- **Its ability to control and store charge leads to the development of MOSFETs and memory devices.**
- **Flash memory devices utilize charge storage in floating gate MOS structures to provide fast, reliable, and non-volatile data storage, making them essential in today's digital world.**

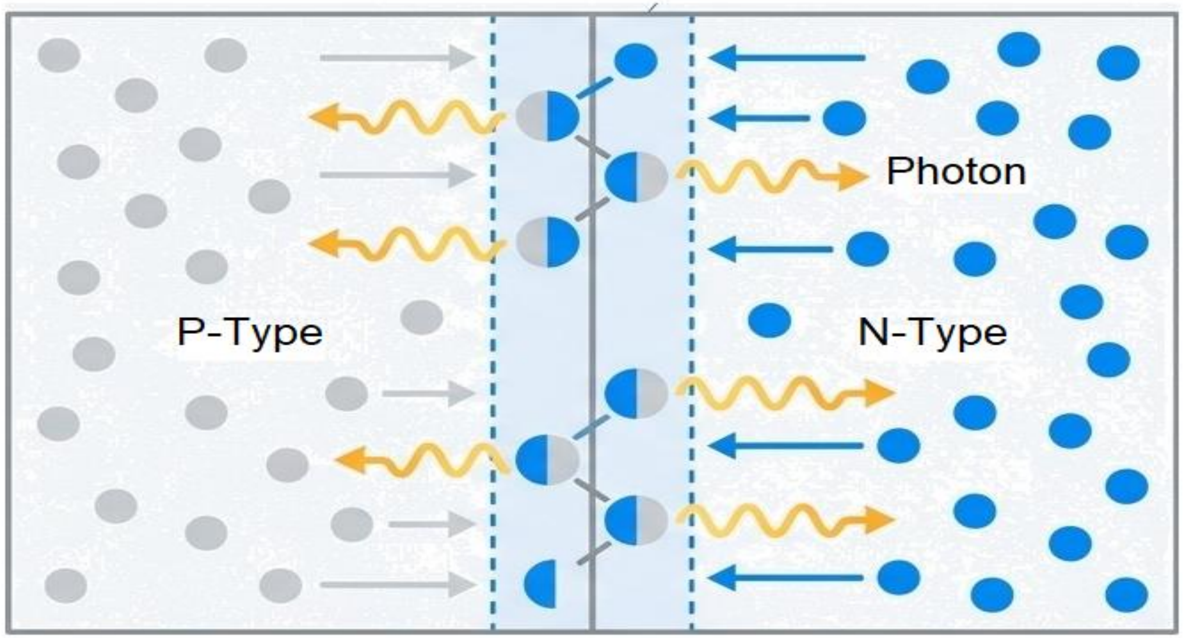
Applications of p-n junction diode

Dr Archana Sharma

From Current to Color: The magic of PN junction

The foundation of modern lighting is the P-N junction diode.

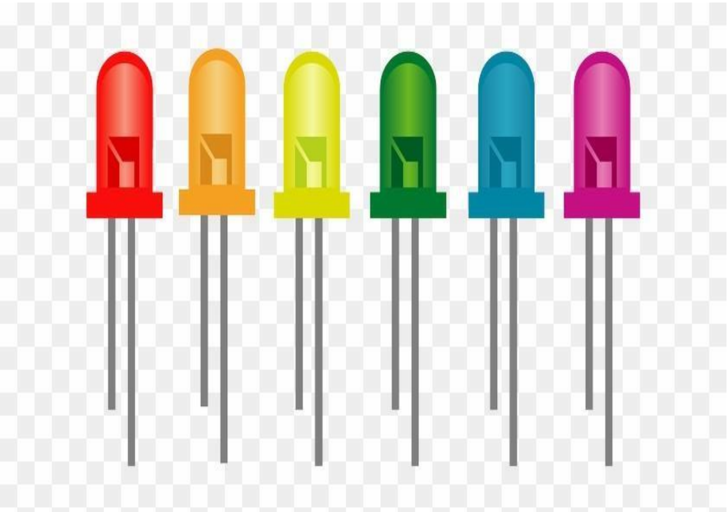
When a current is applied in the forward direction (forward bias), electrons from n-side are pushed toward the junction, where they meet holes from the P-side. This 'recombination' of charge carriers releases energy.

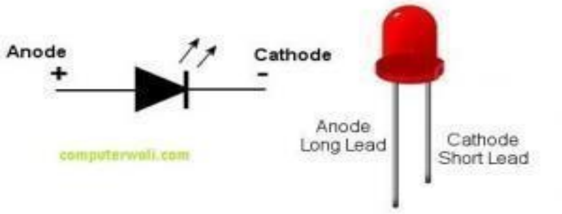


In specific semiconductor materials, this energy is emitted as photons particles of light.

LED: The Efficient Light Source

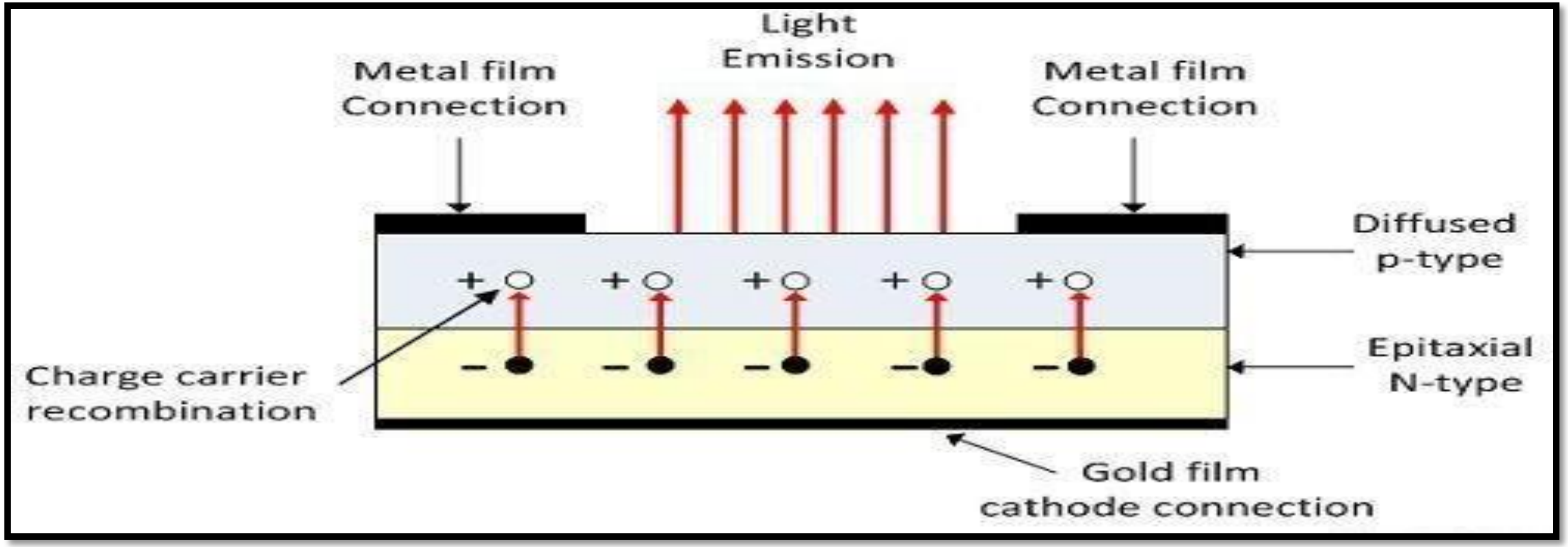
- The LED is a PN-junction diode which emits light when an electric current passes through it in the forward direction.
- In the LED, the recombination of charge carrier takes place.
- The electron from the N-side and the hole from the P-side are combined and gives the energy in the form of heat and light.
- The LED is made up of semiconductor material which is colourless, and the light is radiated through the junction of the diode.
- The LEDs are extensively used in segmental and dot matrix displays of numeric and alphanumeric character.
- The several LEDs are used for making the single line segment while for making the decimal point single LED is used.





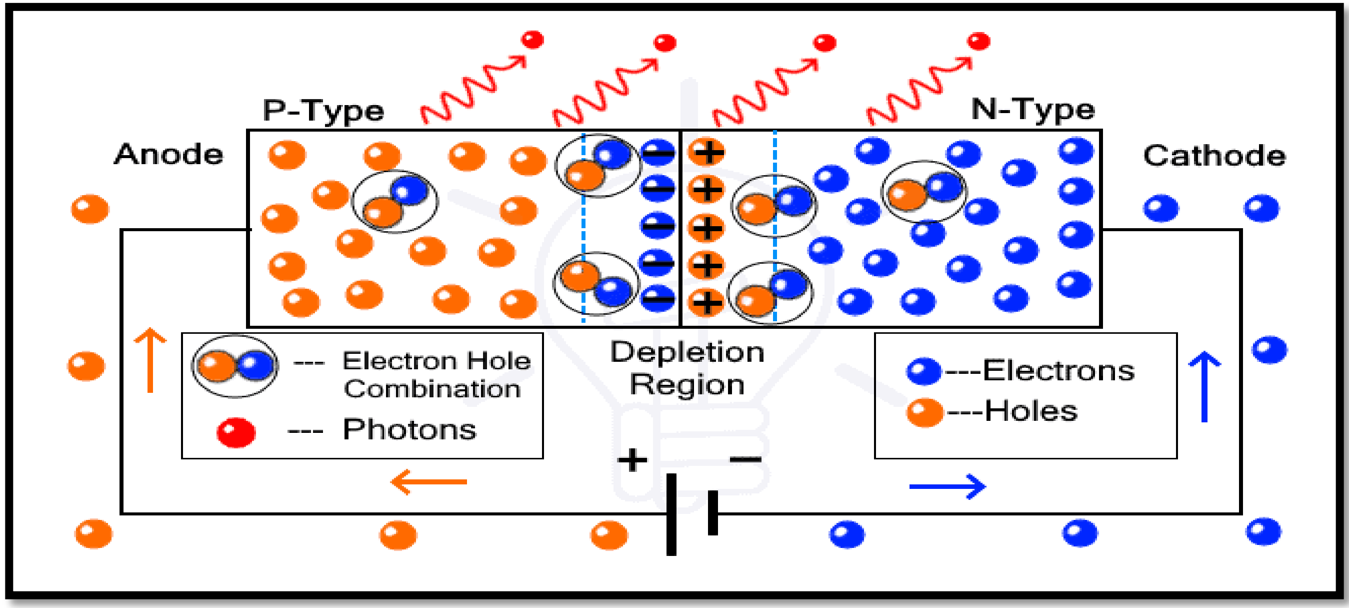
Construction of LED

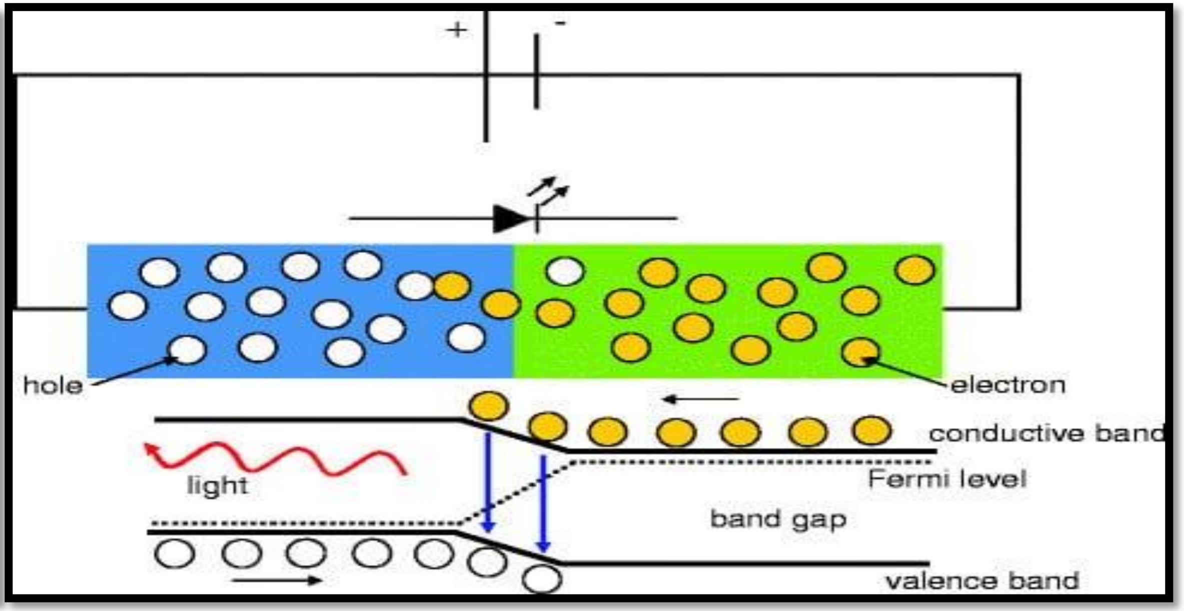
- The recombination of the charge carrier occurs in the P- type material and hence P-material is the surface of the LED.
- For the maximum emission of light, the anode is deposited at the edge of the P-type material.
- The cathode is made of gold film. This gold layer of cathode helps in reflecting the light to the surface.
- The gallium arsenide phosphide is used for the manufacturing of LED which emits red or yellow light for emission.
- The LED are also available in Green, Yellow, Amber and Red in colour.



Working of LED

- LED works only in forward bias condition. The free electrons from n-side and the holes from p-side are pushed towards the junction.
- The recombination of e-h takes place in the depletion region → the width of depletion region decreases → more charge carriers will cross the p-n junction.
- Recombination takes place in depletion region as well as in p-type and n-type semiconductor.
- The free electrons in the conduction band releases energy in the form of light before they recombine with holes in the valence band.





Advantages of LED's

- The brightness of light depends on the current flowing through the LED. Hence, the brightness can be easily controlled by varying the current.
- Light emitting diodes consumes low energy, are very cheap, light in weight and readily available.
- Smaller size, longer lifetime, operates very fast., can be turned on and off in very less time.
- Do not contain toxic material.

Disadvantages of LED's

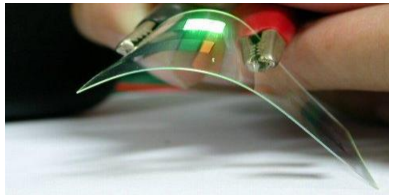
- LEDs need more power to operate than junction diodes.
- Luminous efficiency of LEDs is low.

Applications of LED's

- Burglar alarms systems
- Calculators
- Picture phones
- Traffic signals
- Digital computers
- Aviation lighting, Multimeters
- Microprocessors
- Digital watches
- Automotive heat lamps
- Camera flashes

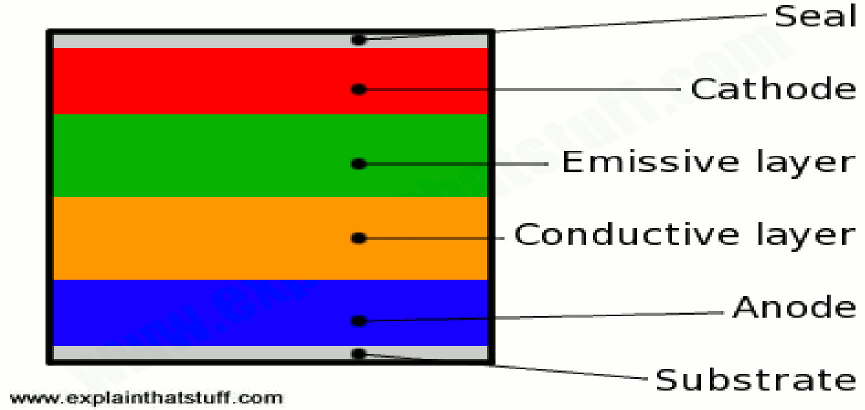
OLED's: Flexible, Thinner, and Brighter Displays

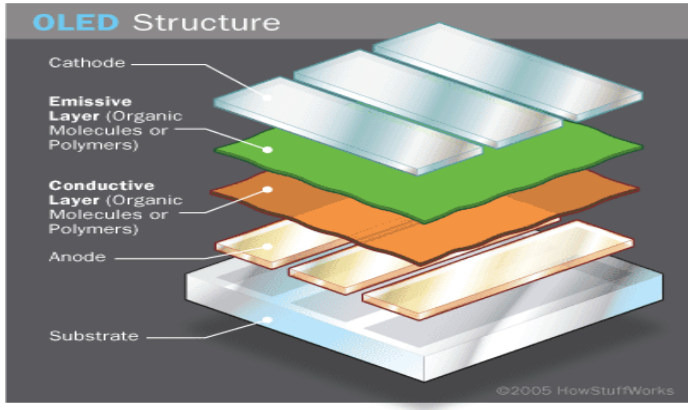
The Shift to Organic: Organic Light Emitting Diodes (OLEDs) replace inorganic semiconductor crystals with thin films of organic compounds that exhibit electroluminescence.



Construction of an OLED

- Layered Structure: An OLED is a stack of fine layers
- Substrate: The base which can be a flexible plastic.
- Anode & Cathode: Supply voltage across the device.
- Conductive Layer: Transports "holes" from the anode.
- Emissive Layer: Transports electrons from the cathode.
- Recombination of electrons and holes in this layer emits light.





The Decisive Advantage: The key benefit of OLEDs is that they are self-emissive, each pixel is its own light source. This eliminates the need for a backlight, allowing for perfect blacks, higher contrast, thinner profiles, and flexible displays.

Advantages

- Organic layers present in an OLED are better than the crystalline layers in LED or LCD as they are flexible, lighter, and thinner than crystalline layers.
- Due to the presence of organic layers, OLEDs are brighter than LEDs.
- OLEDs consume less power than LCD and LEDs.
- They have faster response time and better picture quality and viewing angle than LEDs.

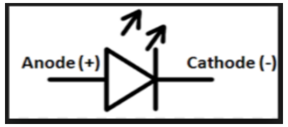
Disadvantages

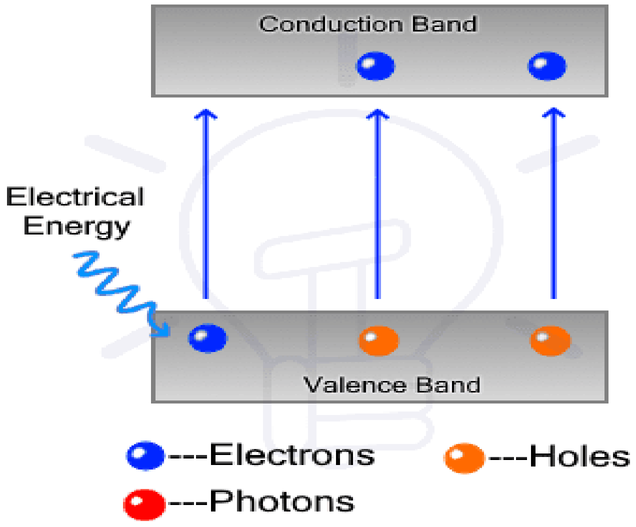
- Currently, the biggest drawback of an OLED is its lifetime.
- The Red and Green OLED films have a lifetime ranging from 46, 000 to 230, 000 hours while Blue organics have a shorter lifetime of around 14, 000 hours.
- The process of manufacturing an OLED is expensive right now.
- They are easily damaged by water.

OLEDs are currently used in smartphones, television, digital cameras, computer monitors, portable gaming consoles etc.

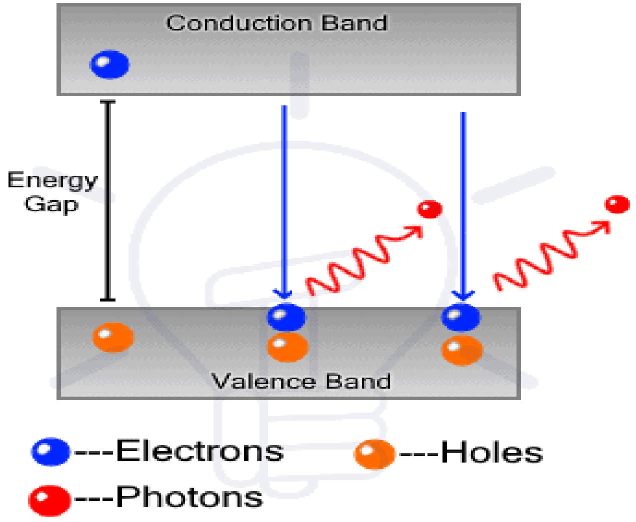
LASER DIODES: Amplifying Light into a Coherent Beam

- It is a semiconductor-based PN junction device that converts electrical energy into light energy similar to LED.
- The Key Difference: While LEDs rely on spontaneous emission—random, incoherent light—Laser Diodes use stimulated emission.
- An initial photon triggers an excited electron to release an identical photon, creating an amplified, coherent, and monochromatic beam.

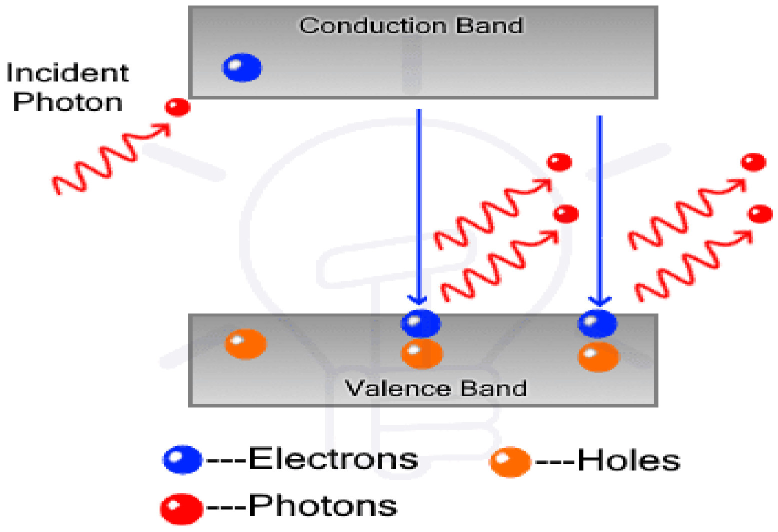




Energy Absorption



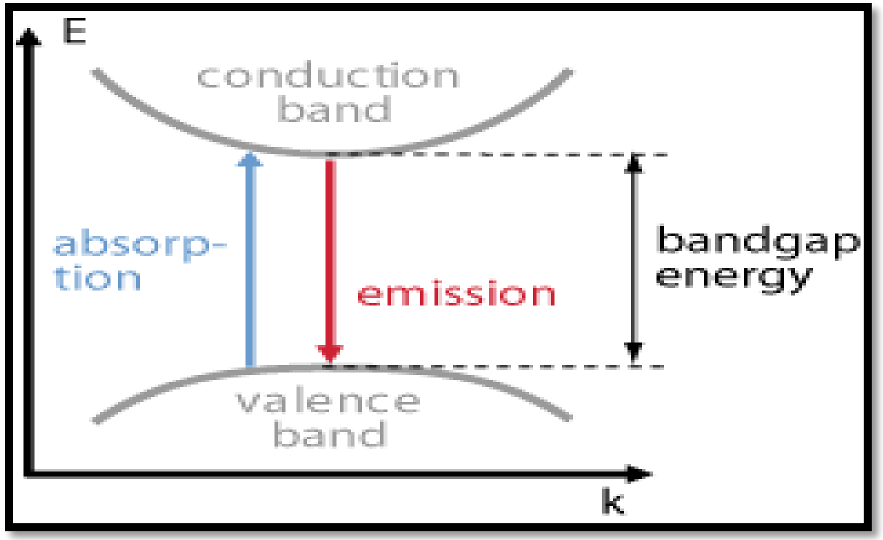
Spontaneous Emission



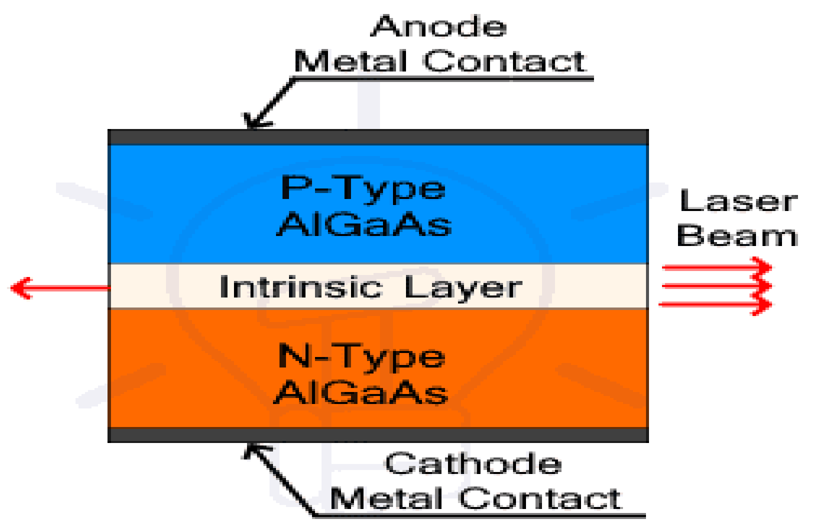
Stimulated Emission

Construction of a laser diode

Laser diode construction requires the emitting layer to be a direct bandgap semiconductor material with a structure that enables direct emission by recombination of carriers.

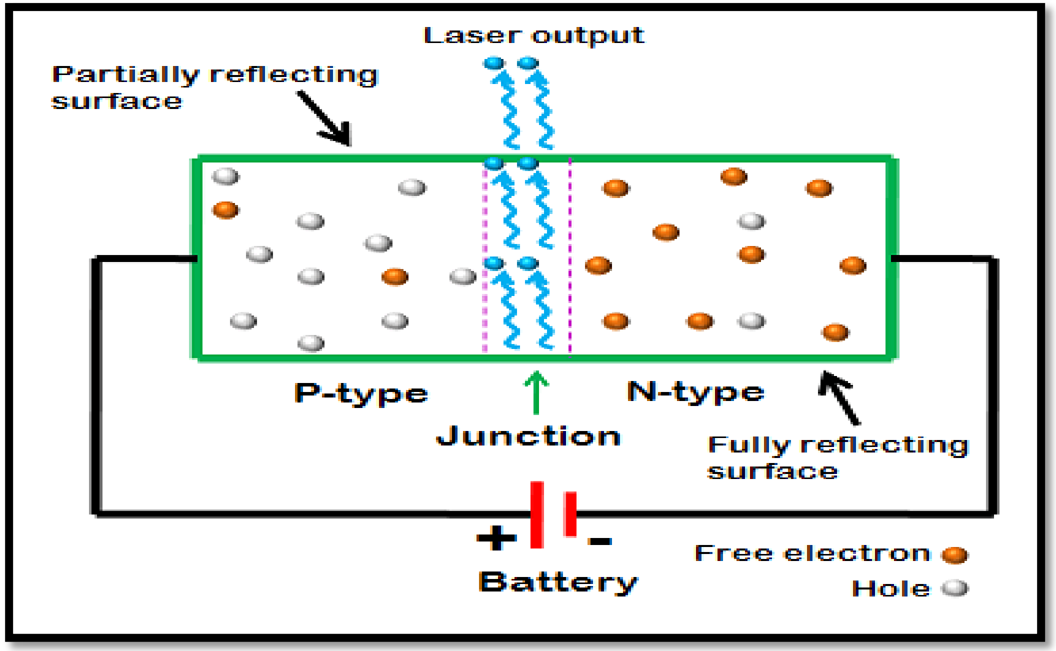


- It's made of three semiconductor layers P-type, N-type and intrinsic layer to form a PIN structure.
- The semiconductor material used is GaAs with trivalent and pentavalent impurities for p and n-type, while the intrinsic layer is made of pure GaAs layer → homo-junction laser diode since it uses the same material.
- The intrinsic layer is sandwiched between the P-type and N-type layer.



Laser Diode Construction

- Metal plates connected with p-layer and n-layer forms the terminals anode (+) and cathode (-) respectively.
- The light/photons produced in the intrinsic layer → called active layer.
- Intrinsic layer increases the width of the PN junction → increases the number of electron-hole recombination → increases the generation of photons.



- The active region is covered with semi-reflective material to collect all photons and redirect them in one direction.
- Furthermore, there is a lens that focuses the light even more into a single spot of a high-power beam. The whole structure is covered in a metal casing with a single opening for light emission.

Advantages

- Laser diode is very small in size and compact.
- It has high efficiency.
- The laser output can be easily increased by increasing the junction current.
- It is operated with less power than ruby and CO_2 lasers.
- It requires very little additional equipment.
- It emits a continuous wave output or pulsed output.

Disadvantages

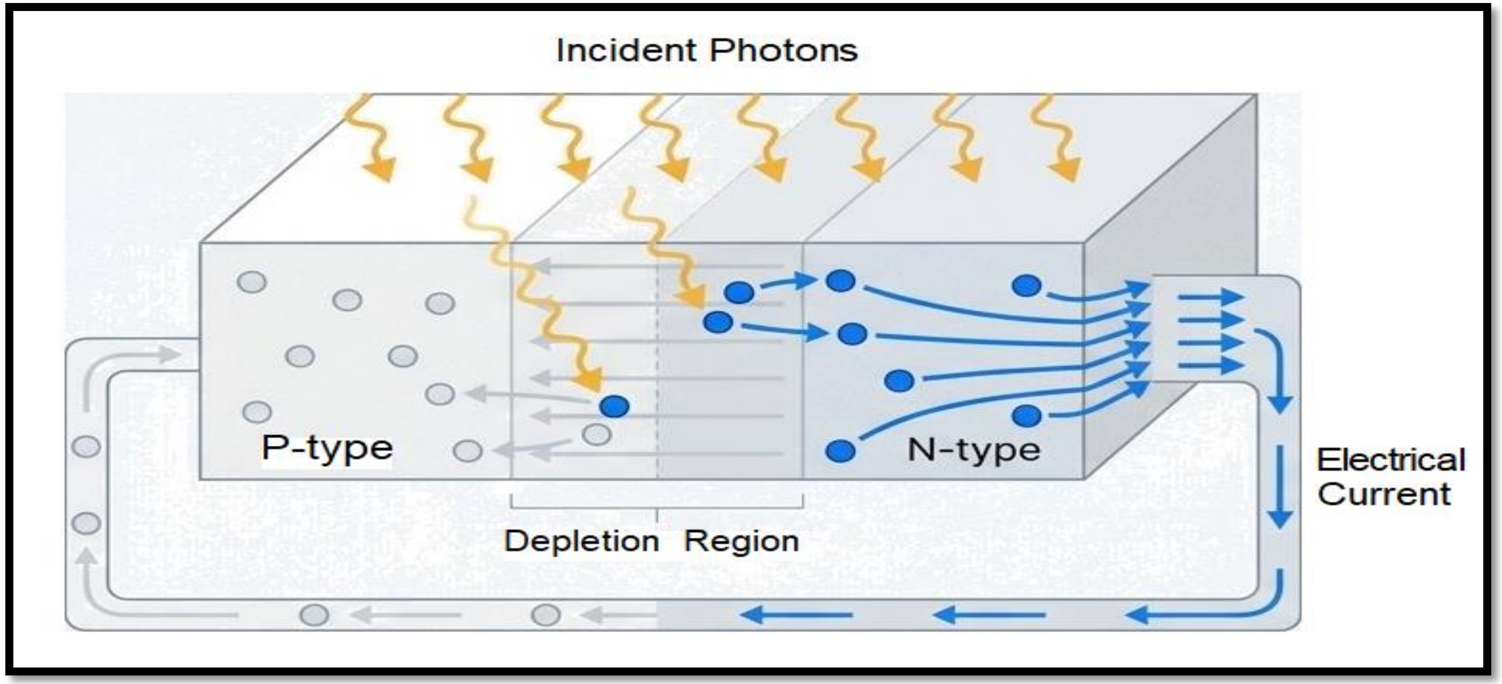
- Laser output beam has large divergence.
- The purity and monochromaticity are poor.
- It has poor coherence and stability.

Applications of Laser Diode

- Used in fibre optic communication.
- Used in various measuring devices such as range finders, bar-code readers.
- Used in printing industry both as light sources for scanning images and for resolution printing plate manufacturing.
- Infrared and red laser diodes are common in CD players, CD-ROM and DVD technology. Violet lasers are used in HD-DVD and Blue-ray technology.
- High power laser diodes are used in industrial applications such as heat treating, cladding, seam and for pumping other lasers.

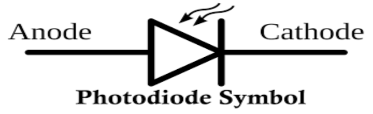
The Other Side of optoelectronic devices: From Photon to Current

- The interaction between light and electricity is a two-way street.
- Just as current can create light, light can create current.
- When a photon with sufficient energy strikes a semiconductor, it can excite an electron, creating a free electron and a "hole."
- This is the foundational principle of light detection and solar power.



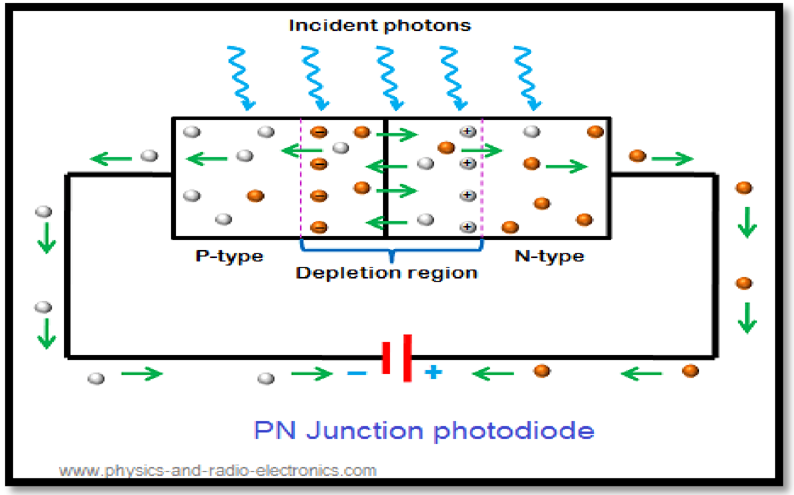
PHOTODETECTORS: The Eyes of Electronics

- An optoelectronic device that is used to detect the incident light or optical power to convert it into an electrical signal.
- Examples of photodetectors are phototransistors and photodiodes.



Working Principle

- **Incident Light: When light (photons) falls on the active region of the photodetector (usually a semiconductor like silicon), the energy from the photons is absorbed by the material.**
- **Generation of Electron-Hole Pairs: The absorbed photon energy excites electrons from the valence band to the conduction band, creating electron-hole pairs.**



- **Separation of Charges: In a photodiode (most common photodetector), a p-n junction is used. The built-in electric field at the junction separates the electrons and holes, causing them to move towards opposite electrodes.**
- **Current Flow: The movement of electrons and holes generates a photocurrent, which is proportional to the intensity of the incident light.**

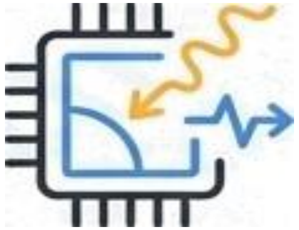
Types of Photodetectors

Photodiodes:

The essential light sensor. A p-n junction where absorbed photons generate electron-hole pairs, which are separated by the junction's electric field to produce a photocurrent proportional to light intensity.

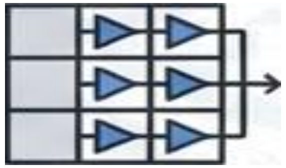
Phototransistors

For high-sensitivity applications. These devices use a transistor structure to amplify the initial photocurrent generated by light, allowing for the detection of very low light levels.



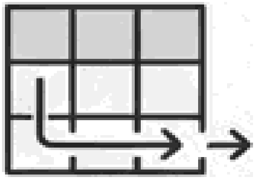
CMOS Image sensors

Complementary metal-oxide-semiconductor (CMOS) sensors can read multiple pixels simultaneously, allowing for faster readout. Integrates amplification and readout circuitry into each pixel, allowing for parallel readout. Known for speed, low power consumption, and high integration



CCD Image sensors

These are image sensors composed of an array of pixels (tiny capacitors). They capture light as charge in each pixel, then transfers the charge sequentially across the array to be read out. Known for high sensitivity and low noise



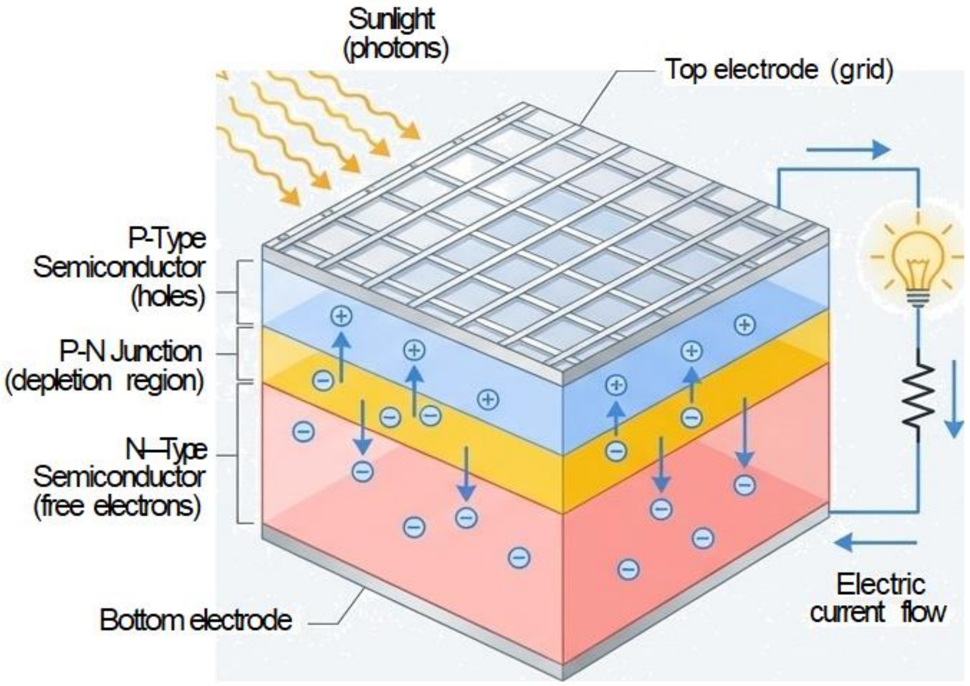
SOLAR CELLS: Powering Our World with the Photovoltaic Effect

A **solar cell or photovoltaic cell** is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect which is a physical and chemical phenomenon.

From Detection to Generation: A solar cell is essentially a very large-area photodiode, engineered not to just detect light, but to efficiently convert its energy into electrical power.

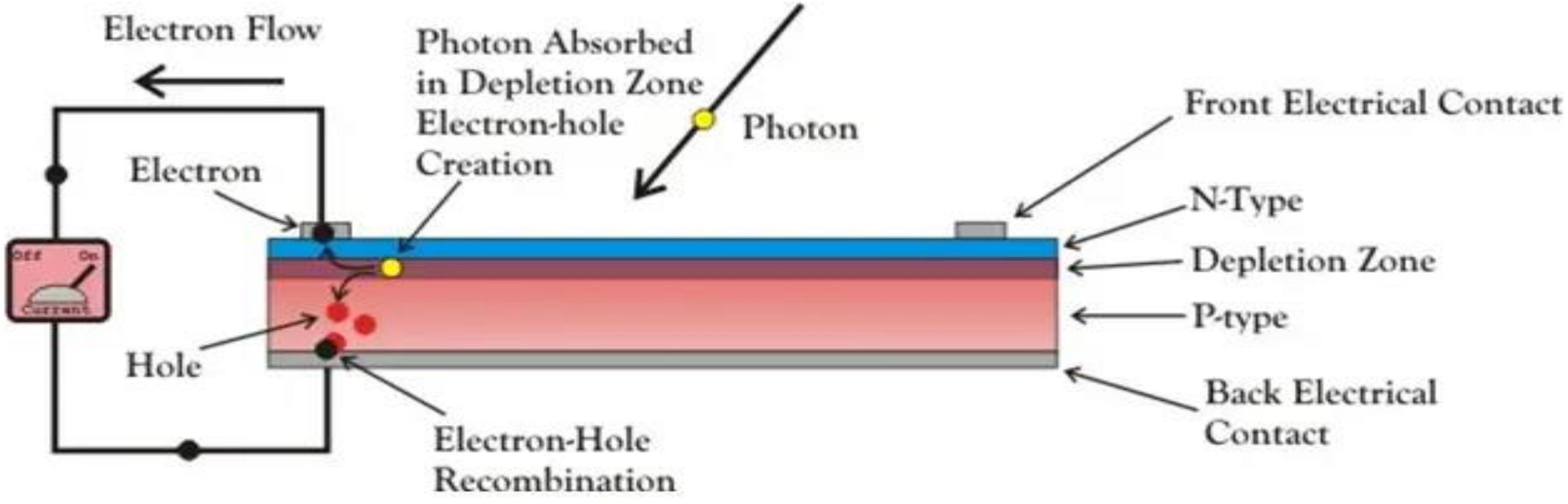
How it Works: Photons strikes the SC material and creates e-h pairs. The built-in electric field of the p-n junction acts as a separator, sweeping electrons to the N-side and holes to the P-side. This accumulation of charge creates a "photovoltage" across the cell, which can drive a current through an external circuit.

Optimized Construction: To maximize light absorption near the junction where charge separation is most efficient, a solar cell is built with a very thin P-type layer grown on top of a much thicker N-type layer. A grid of fine metal electrodes on the surface collects the current without blocking too much light.

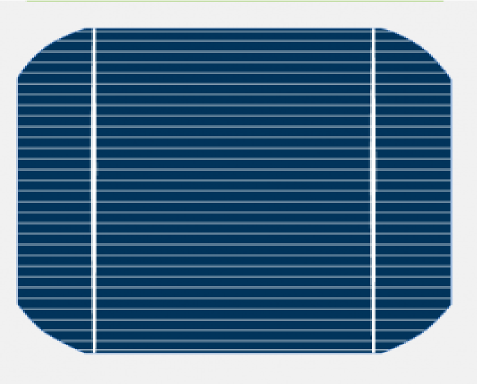


Construction of a solar cell

- A solar cell functions similarly to a junction diode, but its construction differs slightly from typical p-n junction diodes.
- A very thin layer of p-type semiconductor is grown on a relatively thicker n-type semiconductor. We then apply a few finer electrodes on the top of the p-type semiconductor layer.
- These electrodes do not obstruct light to reach the thin p-type layer. Just below the p-type layer there is a p-n junction. A current collecting electrode is also provided at the bottom of the n-type layer. We encapsulate the entire assembly by thin glass to protect the **solar cell** from any mechanical shock.



Advantages of Solar Cell

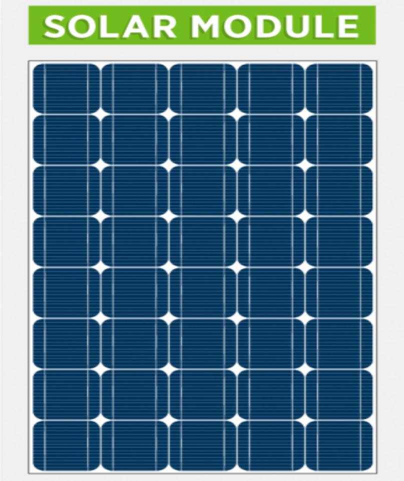


- No pollution associated with it.
- It must last for a long time.
- No maintenance cost.

Disadvantages of Solar Cell

- It has high cost of installation.
- It has low efficiency.
- During cloudy day, the energy cannot be produced and also at night we will not get solar energy.

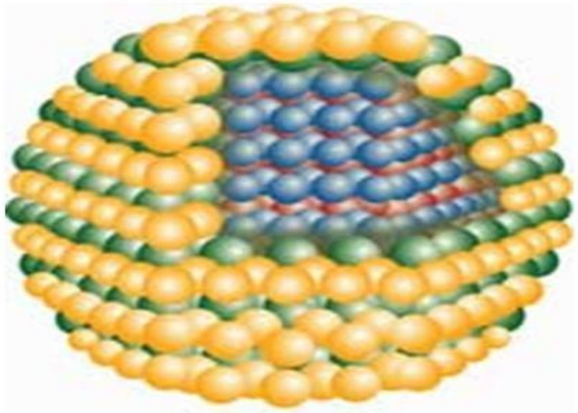
Uses of Solar Generation Systems



- It may be used to charge batteries.
- Used in light meters.
- It is used to power calculators and wrist watches.
- It can be used in spacecraft to provide electrical energy.

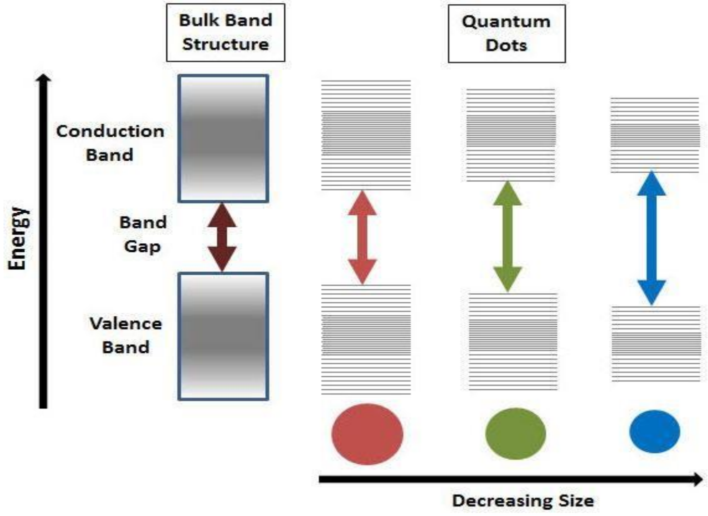
QUANTUM DOTS: Where size determines colours

- **What They Are**: quantum dots are semiconductor nanocrystals so small (2-10 nanometers) that their electronic properties are governed by quantum mechanics. They are essentially "artificial atoms".



- **The Quantum Confinement Effect**: The core principle is "quantum confinement." The physical size of the dot dictates its energy band gap.
- A smaller dot has a larger band gap, requiring more energy to excite and releasing more energy (a higher frequency photon) when it returns to its ground state.

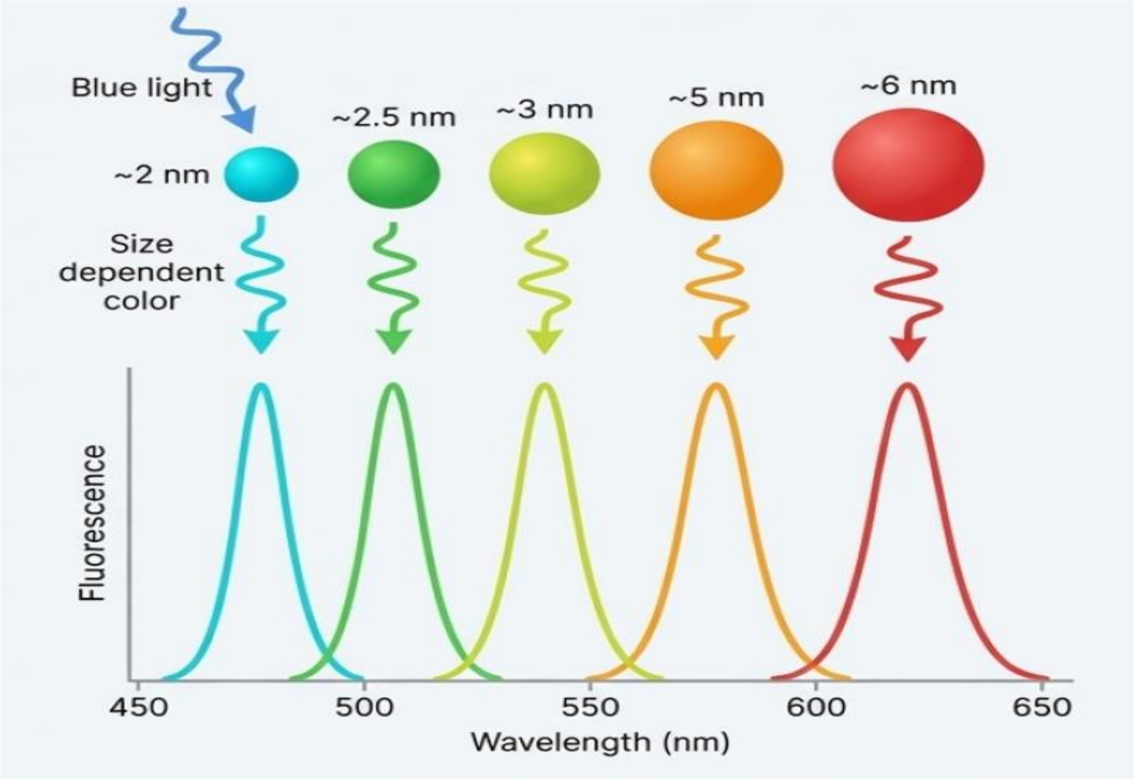
- **The Tunable Result**: This creates a direct, predictable relationship between size and color.
- Smaller dots (e.g., 2 nm) emit higher-energy, shorter- wavelength light (Blue/Green).
- Larger dots (e.g., 6 nm) emit lower-energy, longer- wavelength light (Orange/Red).



This allows for the creation of extremely pure and precisely tuned colours.

Role of quantum dots in display

- Typically, in display applications QDs convert incident blue light into red or green; the output colour depends on the size of the quantum dot.
- Although they have several other potential applications, the first major use of these materials has been in display devices.

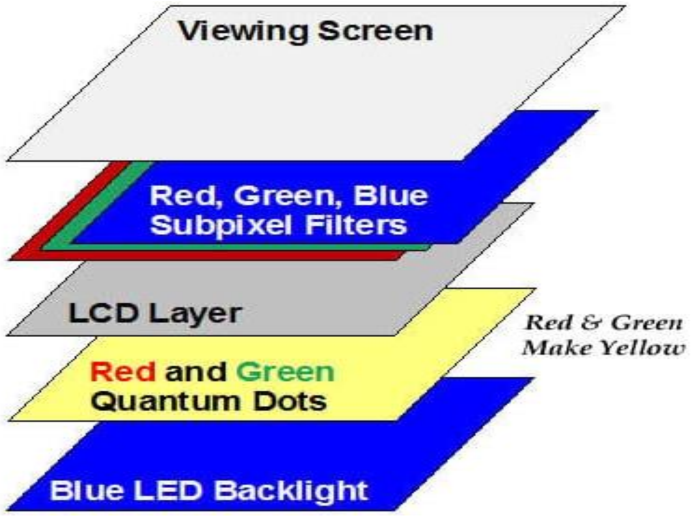


Construction of a QLED

<u>How It Works (Photoluminescence)</u>: In current QLED displays, quantum dots act as advanced color filters. They don't generate light from electricity directly; instead, they absorb light from one source and re-emit it at a different, highly specific color.

<u>The QLED Panel Structure</u>

- **LED Backlight**: A powerful and efficient array of blue LEDs provides the initial light source.
- **Quantum Dot Film**: A layer containing precisely engineered red and green quantum dots is placed in front of the backlight.
- **Color Conversion**: The blue light excites the quantum dots. The green dots absorb blue light and emit pure green; the red dots absorb blue light and emit pure red.
- **LCD Matrix**: The resulting pure red, green, and blue light passes through a standard liquid crystal layer that shapes the light into the final image.



This process creates exceptionally pure primary colors, resulting in a significantly wider color gamut, higher peak brightness, ano greater color volume than traditional LCDs.

Quantum Dot Materials

•Quantum dots can be made from a range of materials, currently the most commonly used materials include Zinc Sulphide, Lead Sulphide, Cadmium Selenide and Indium Phosphide.

•Cadmium selenide is commonly used in QLED and Solar Cells.

Qdots	Conduction Band (eV)	Valence Band (eV)	Particle Size (nm)
CdSe	4.4	6.5	5
CdSe/CdS	4.4	6.5	4.6
CdSe/CdS	4.7	6.8	4
CdSe/CdS/ZnS	4.8	6.8	6.8
CdSe/ZnS	4.4	6.5	
CdSe/ZnS	4.3	6.5	
CdSe/ZnS	4.8	6.5	
CdSe/ZnS	4.6 (CdSe)	6.8 (CdSe)	5.8
CdSe/ZnS	4.7	6.7	
CdSe/ZnS/CdS	3.9	6.0	3–8.3

QLED technology's applications

- **Televisions & Monitors**
- **Smartphones & Tablets**
- **Gaming & AR/VR**
- **Professional & Industrial**
- **Digital Signage**
- **Specialized Lighting**
- **Control Rooms & Surveillance**
- **Medical Imaging:**
- **Photomedicine**
- **Flexible Displays**
- **Wearable Devices**
- **Content Creation**